
STACK OVERFLOW DEVELOPER SURVEY 2024

Technology Trends, Job Market & Developer Demographics

Data Analytics Project

Ashwini Wadkar
September 2026

From developer survey insights
to data-driven decisions



OUTLINE

- 01  Executive Summary
- 02  Introduction
- 03  Methodology
- 04  Results – Visualization (Charts) & Dashboard
- 05  Discussion – Findings & Implications
- 06  Conclusion



EXECUTIVE SUMMARY

KEY INSIGHTS AT A GLANCE



Job Demand

C has the highest number of job postings: 25,114.



Salary

Swift has the highest annual average salary: \$130,801.



Programming Trends

JavaScript leads current usage; JavaScript, SQL and TypeScript show strong future preference.



Database Trends

PostgreSQL leads current usage (25,910) and future preference (26,904).

DEMOGRAPHICS • Ages 25–34 = 41.3% of respondents • Bachelor's degree holders = largest education group

INTRODUCTION

WHY THIS ANALYSIS MATTERS



Purpose

Analyze developer technology trends, job demand, salary patterns, future technology preferences, and respondent demographics.



Target Audience

Technology recruiters, hiring managers, and organizations planning future technology skills.



Value

Identify in-demand technologies and emerging preferences to support data-driven technology and workforce decisions.

METHODOLOGY

HOW THE ANALYSIS WAS DONE

01 DATA COLLECTION

- Stack Overflow Developer Survey
- Job API → Job-posting data Analysis
- Web scraping → salary data Analysis

02 WRANGLING

- SQL for data querying & extraction
- Python & Pandas for data cleaning
- Handle missing values and duplicates
- Cleaning, transformation and normalization

03 EDA

- Analyze Distributions and outliers
- Examine Relationships and correlations
- Calculate Summary statistics and grouped measures

04 VISUALIZATION

- Matplotlib / Seaborn for Analysis Visualization
- IBM Cognos Analytics for interactive Dashboard
- PowerPoint for presenting findings

RESULTS

KEY FINDINGS OVERVIEW



Job Market

C leads job postings with 25,114 postings.



Salary

Swift has the highest annual average salary at \$130,801.



Technology Trends

JavaScript leads current usage; JavaScript, SQL and TypeScript show strong future preference.



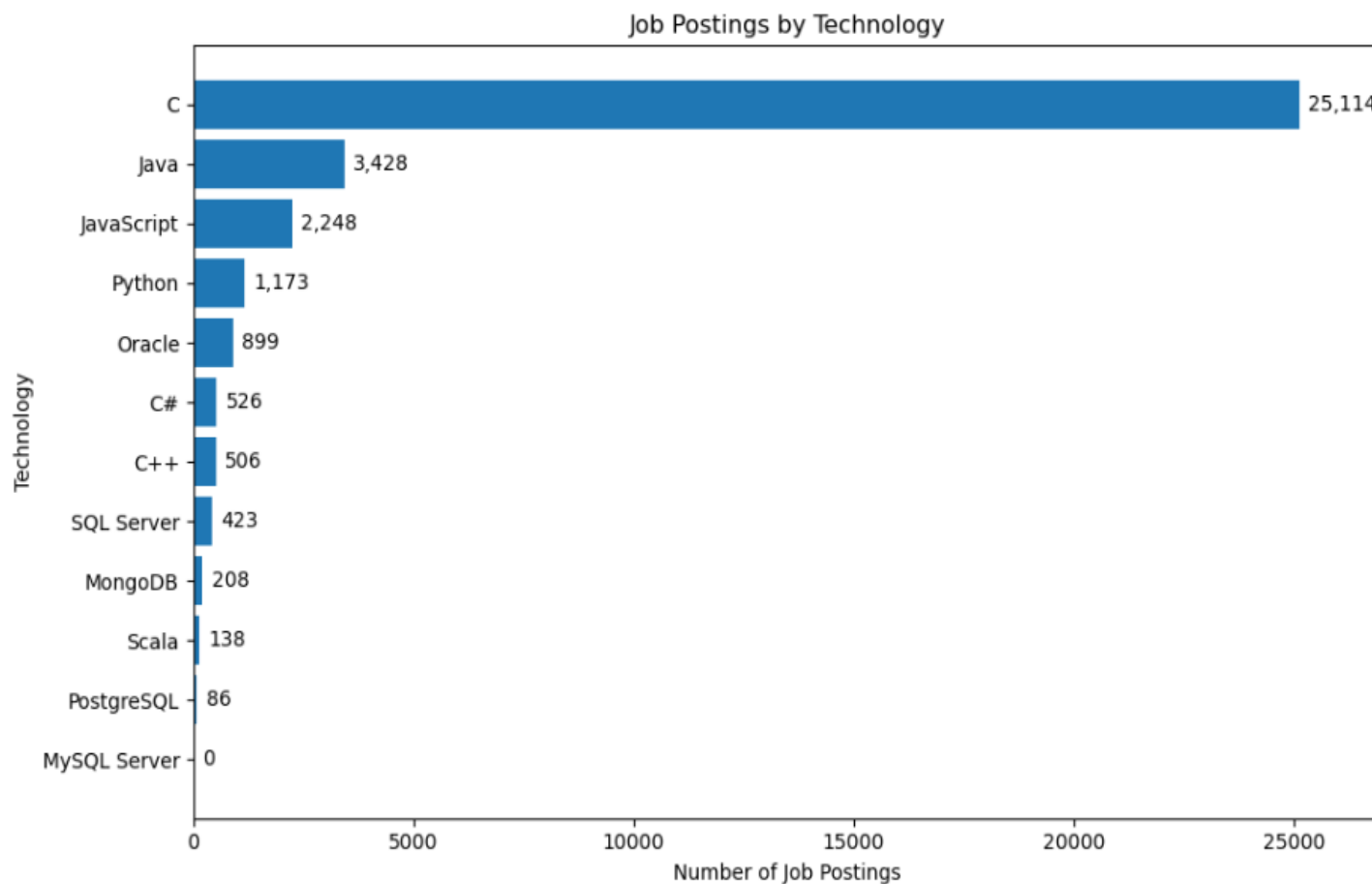
Demographics

Ages 25–34 are the largest group at 41.3%; Bachelor's degree holders are the largest group.

DATA → PATTERNS → INSIGHTS → DECISIONS

JOB POSTINGS

TOP LANGUAGES BY NUMBER OF JOB POSTINGS



KEY TAKEAWAY

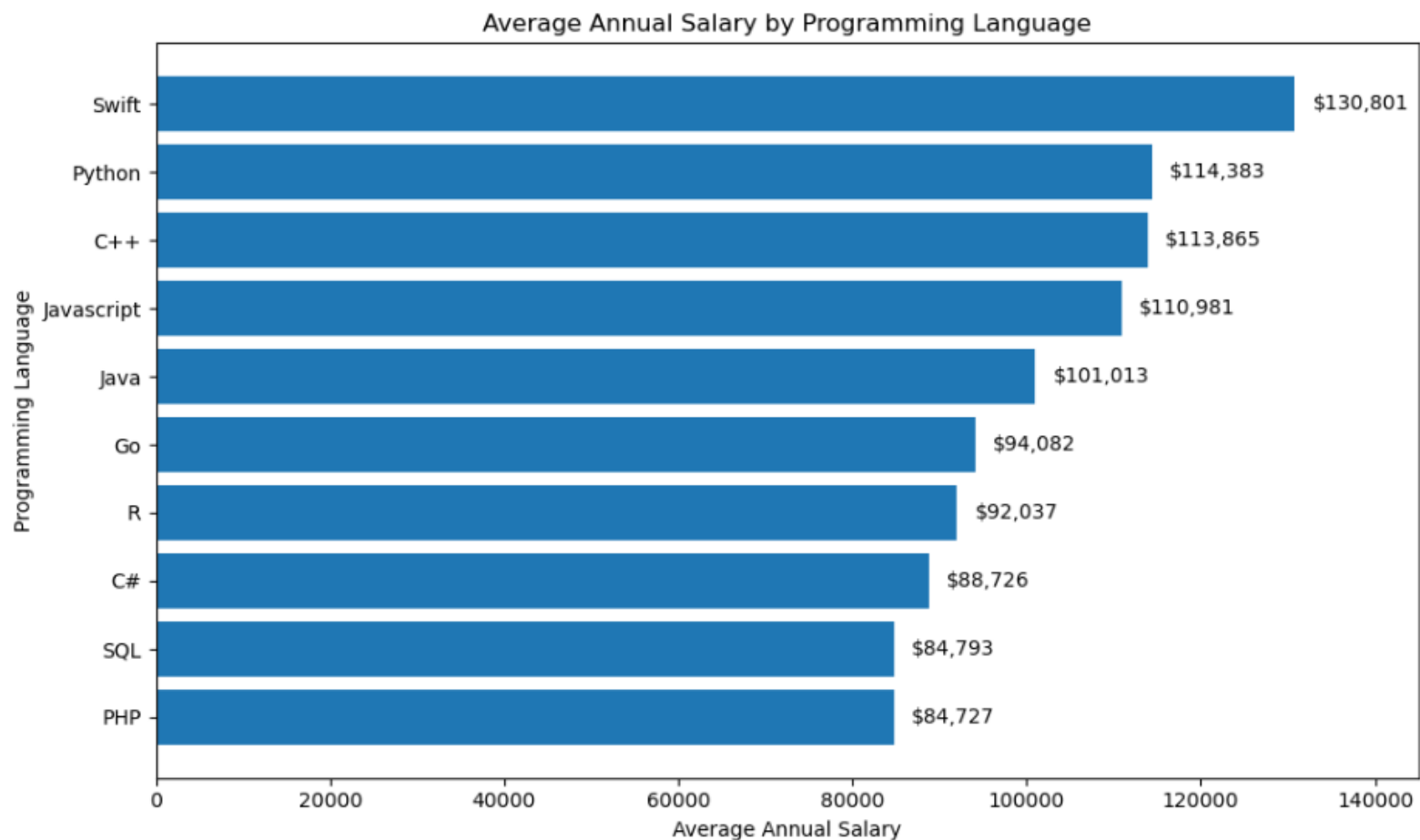
C dominates the job-posting data, while Java, JavaScript and Python show comparatively strong demand.

Source: Job-posting API dataset

C = 25,114

POPULAR LANGUAGES

USAGE, SALARY & INDUSTRY RELEVANCE



WHY IT MATTERS

Several widely used languages, including Python, JavaScript and Java, combine strong salaries with broad industry relevance.

Skill choice should balance market demand + salary + relevance.

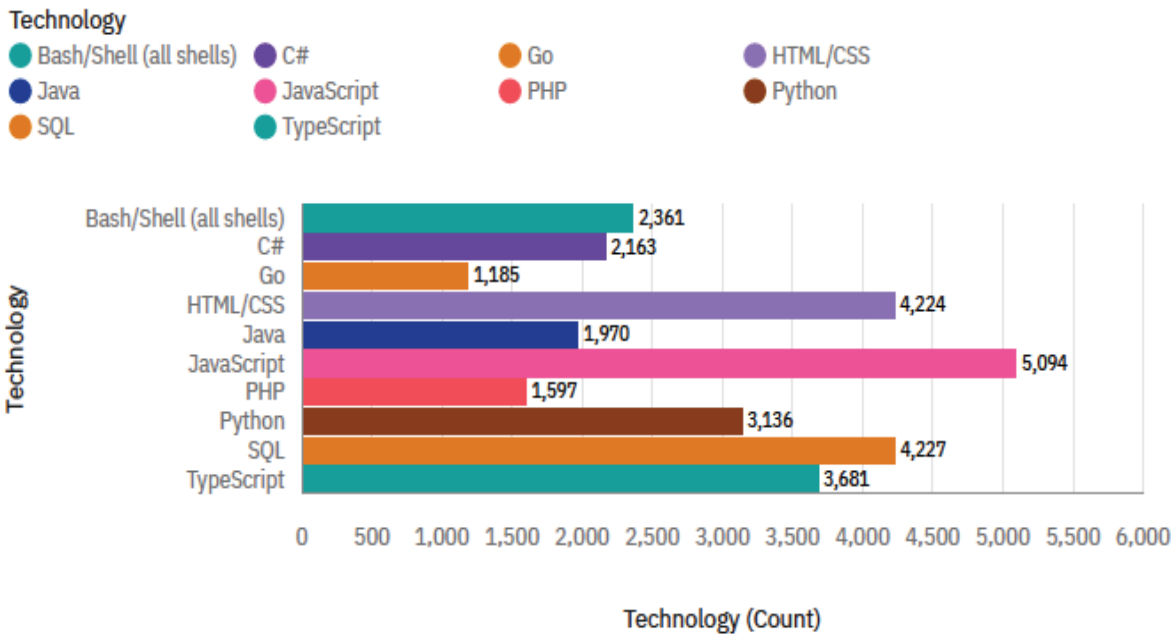
PROGRAMMING LANGUAGE TRENDS

CURRENT USAGE VS. FUTURE PREFERENCE

TOP 10 LANGUAGES DEVELOPERS USED

Current Technology Usage

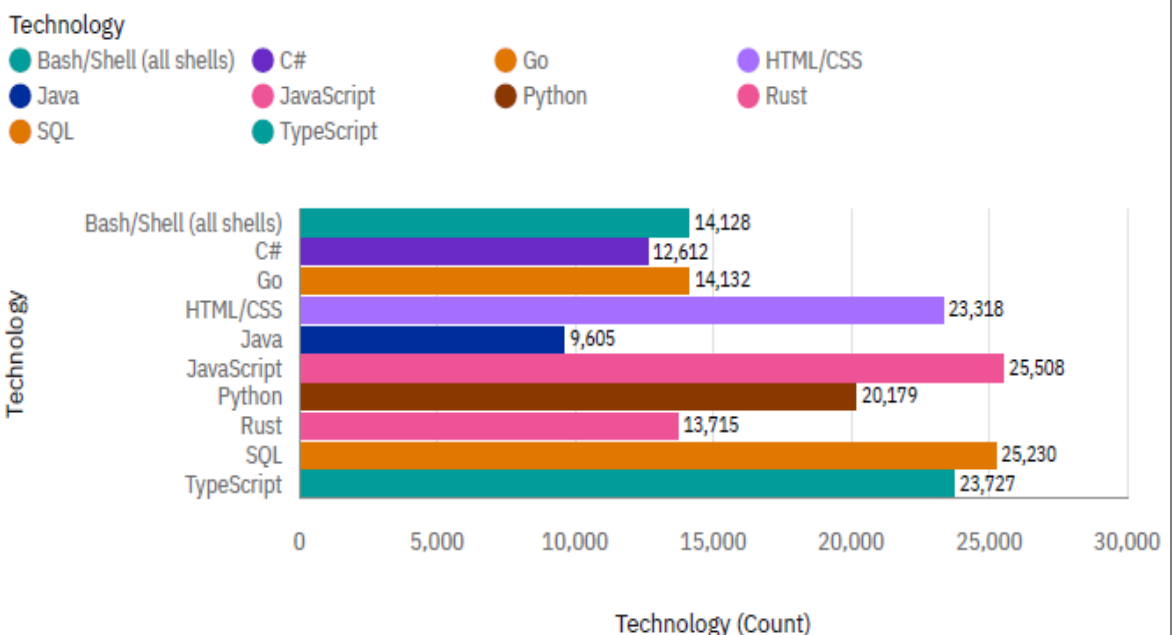
Top 10 Programming Languages Used



TOP 10 LANGUAGES DEVELOPERS WANT TO WORK WITH

Future Technology Trend

Top 10 Languages Developers Want to Work With



PROGRAMMING LANGUAGE TRENDS

FINDINGS & IMPLICATIONS

FINDINGS

- JavaScript leads both current usage and future interest.
- SQL, TypeScript and HTML/CSS show strong future interest.
- Python remains highly relevant across both periods.

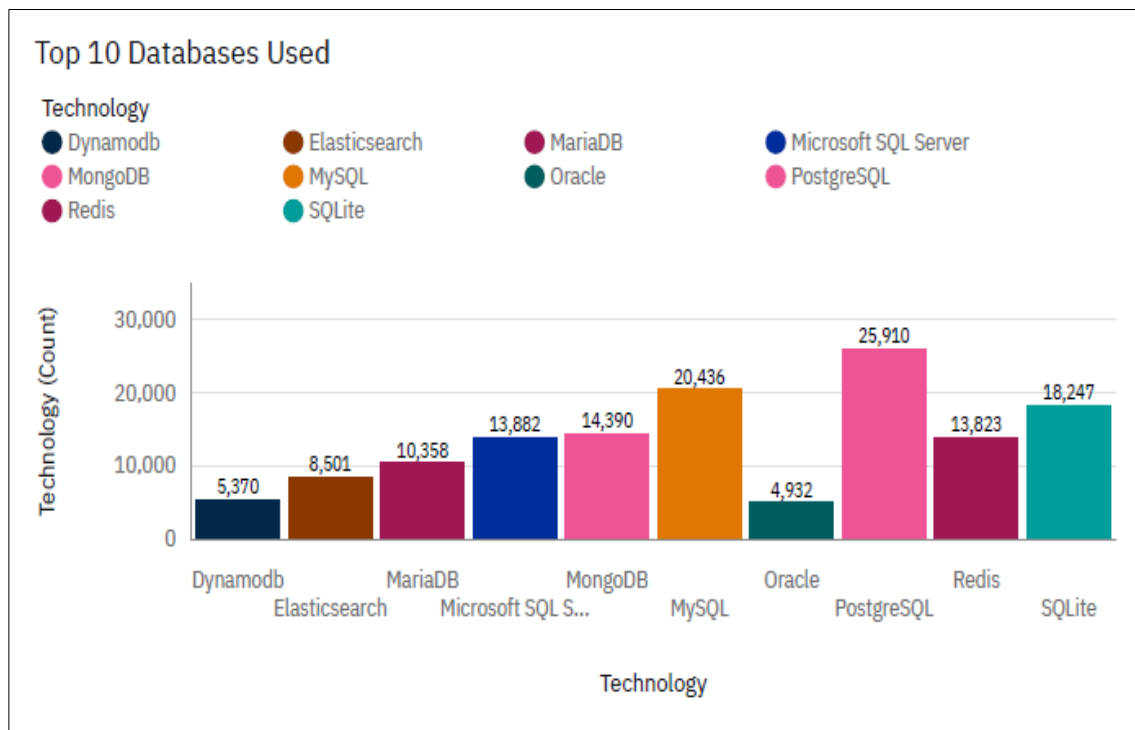
IMPLICATIONS

- JavaScript should remain a key development skill.
- Broader skills across complementary technologies can improve career opportunities.
- Python is a strong long-term language to learn and develop.

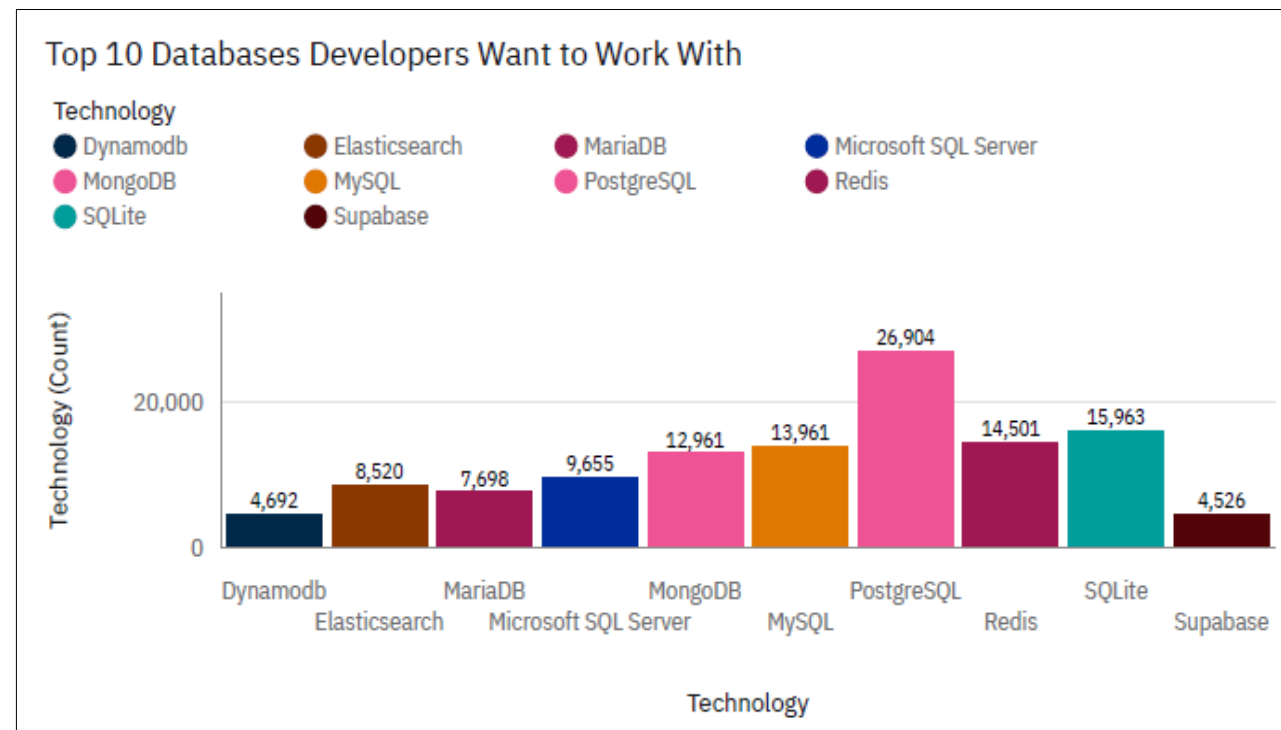
DATABASE TRENDS

CURRENT USAGE VS. FUTURE PREFERENCE

TOP 10 DATABASES USED BY DEVELOPERS



TOP 10 DATABASES DEVELOPERS WANT TO WORK WITH



DATABASE TRENDS

FINDINGS & IMPLICATIONS

FINDINGS

- PostgreSQL leads both current usage and future interest.
- Redis and Supabase show growing future interest.
- MySQL has significantly lower future interest than current usage.

IMPLICATIONS

- PostgreSQL shows the strongest future preference among surveyed developers.
- Knowledge of both relational and modern database technologies is valuable.
- Future technology demand should guide database skill development.

DASHBOARD

A UNIFIED VIEW OF KEY INSIGHTS

**The dashboard brings together key insights from
Technology Trends and Developer Demographics.**

TECHNOLOGY

Languages,
databases & tools
currently used

DATABASES

Technologies
developers want
to work with

DEMOGRAPHICS

Age, education &
geographic profile

DASHBOARD TAB 1: CURRENT TECHNOLOGY USAGE

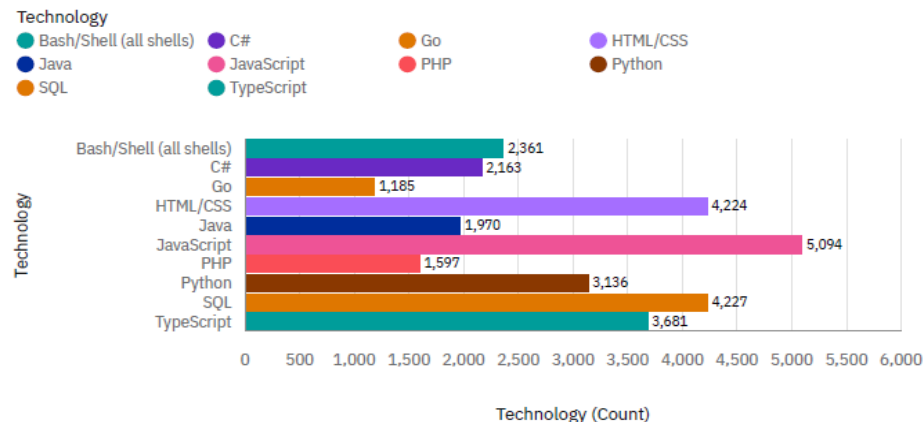
CURRENT TECHNOLOGY USAGE

KEY INSIGHTS

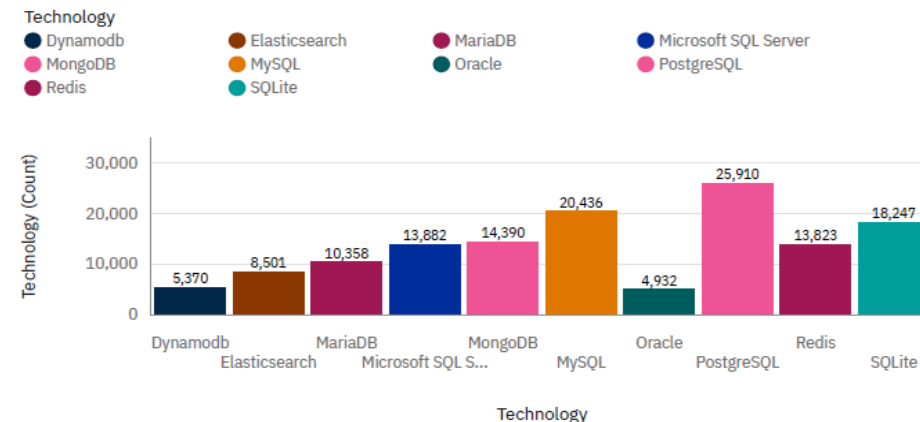
- JavaScript leads current language usage.
- PostgreSQL leads database usage.
- AWS, Node.js and React are prominent technologies.

Current Technology Usage

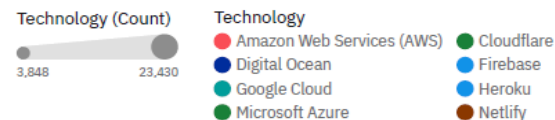
Top 10 Programming Languages Used



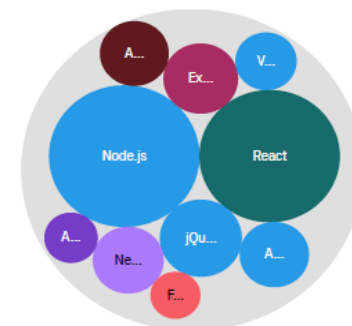
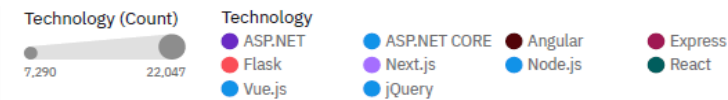
Top 10 Databases Used



Top 10 Platforms Used



Top 10 Web Frameworks Used



DASHBOARD TAB 2: FUTURE TECHNOLOGY TRENDS

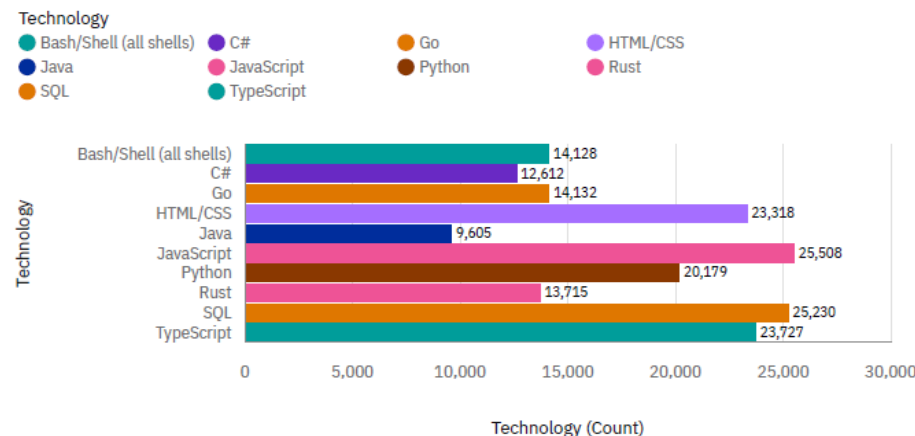
FUTURE TECHNOLOGY TRENDS

KEY INSIGHTS

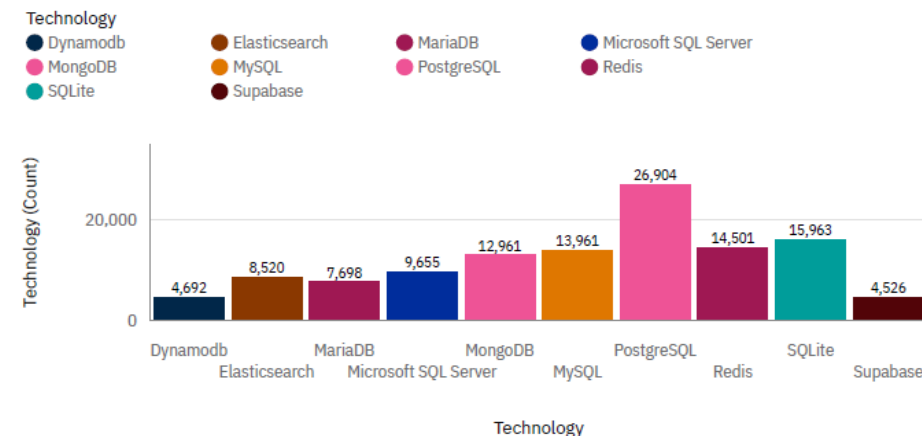
- JavaScript, SQL and HTML/CSS lead future language interest.
- PostgreSQL shows the strongest future database demand.
- AWS, Azure, React and Node.js remain highly preferred technologies.

Future Technology Trend

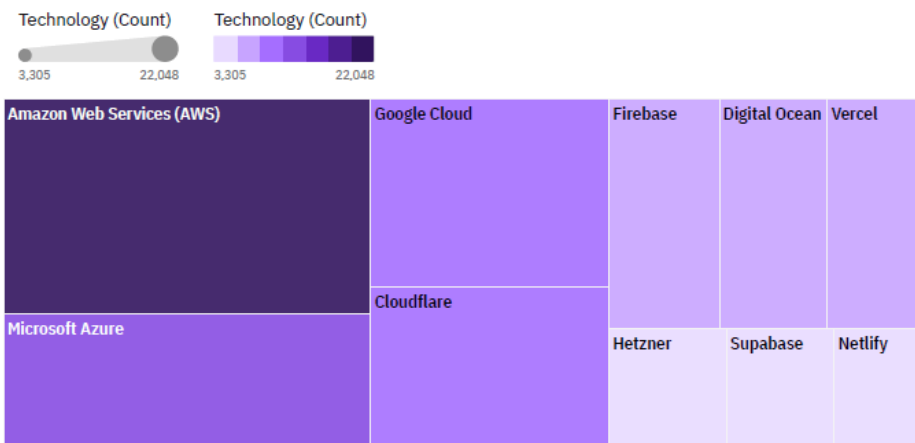
Top 10 Languages Developers Want to Work With



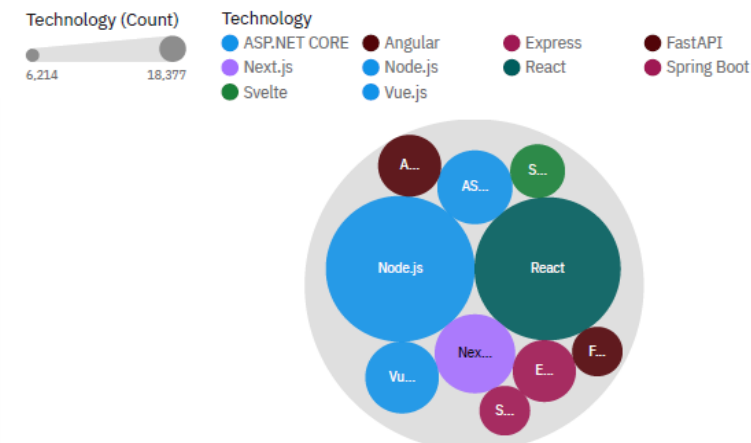
Top 10 Databases Developers Want to Work With



Top 10 Platforms Developers Want to Work With



Top 10 Web Frameworks Developers Want to Work With



DASHBOARD TAB 3: DEMOGRAPHICS

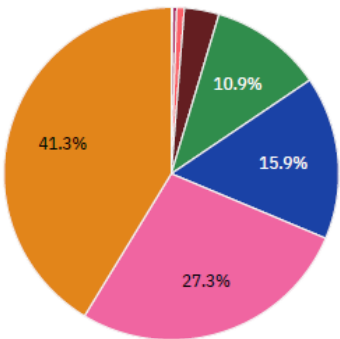
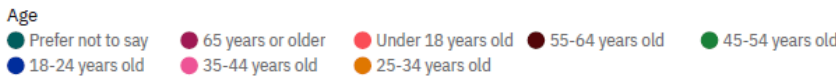
DEMOGRAPHICS

KEY INSIGHTS

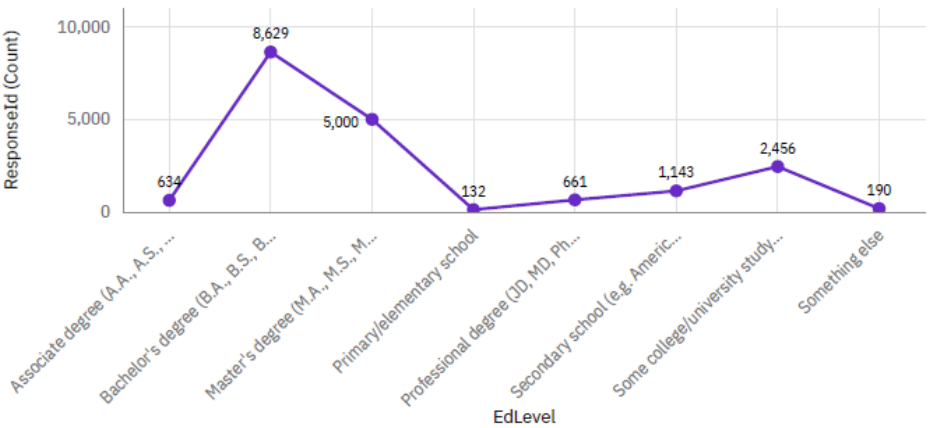
- The 25–34 age group represents the largest share of respondents.
- Bachelor’s degree holders make up the largest education group.
- The survey represents a globally distributed developer community.

Demographics

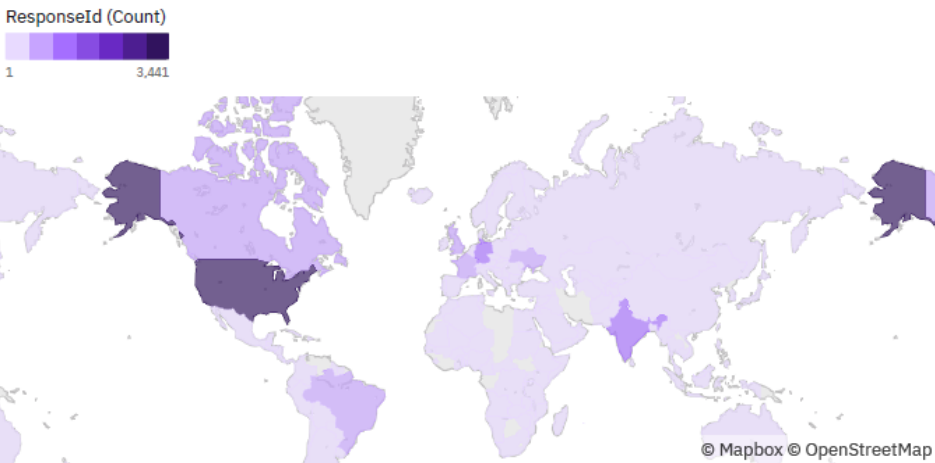
Respondent Distribution by Age



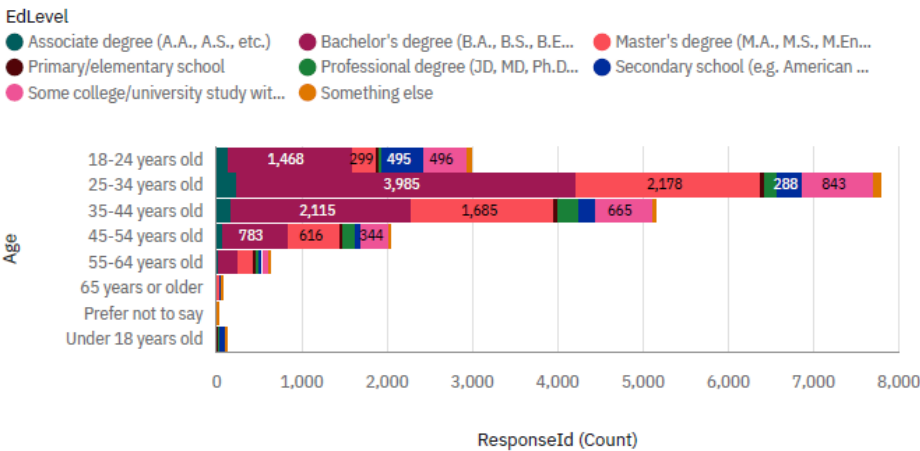
Respondent Distribution by Education Level



Respondent Count by Country



Respondent Count by Age and Education Level



DISCUSSION

INTERPRETING THE FINDINGS



Current Usage

JavaScript, PostgreSQL, AWS and leading web frameworks such as React and Node.js are prominent in current usage.



Future Trends

JavaScript, SQL, PostgreSQL and major cloud/web technologies show strong future interest.



Technology Shift

Comparison between current usage and future preferences highlights opportunities in emerging and evolving technologies.



Developer Profile

Respondent base is largely younger, educated developers, with 25–34 the largest segment and respondents distributed globally.

OVERALL FINDINGS & IMPLICATIONS

FROM DATA TO ACTION

FINDINGS

- JavaScript and PostgreSQL show strong current and future demand.
- Modern web, database and cloud technologies remain highly relevant.
- The respondent population is predominantly young and educated.

IMPLICATIONS

- Prioritize skills aligned with strong current and future demand.
- Continuous upskilling is essential as technology preferences evolve.
- Organizations can use these trends to guide hiring, training and technology planning.

CONCLUSION

KEY TAKEAWAYS

- 1 JavaScript and PostgreSQL remain highly relevant technologies.
- 2 Cloud and web technologies such as AWS, React and Node.js show strong future relevance.
- 3 Continuous upskilling is important as technology trends evolve.
- 4 The analysis provides insights to guide technology learning, hiring and investment decisions.



DATA SOURCES & REFERENCES

FROM DATA TO ACTION

DATA SOURCES

- <https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DA0321EN-SkillsNetwork/labs/module%201/Accessing%20Data%20Using%20APIs/jobs.json> - Stack Overflow Developer Survey 2024
- https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DA0321EN-SkillsNetwork/labs/datasets/Programming_Languages.html - Web-Scraped Salary Dataset
- <https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/n01PQ9pSmiRX6520flujwQ/survey-data.csv> - Stack Overflow Developer Survey 2024
Original source - [Unpacking the 2024 Developer Survey results - Stack Overflow](#)

TOOLS & TECHNOLOGIES

- Python • Pandas • NumPy • SQL • Matplotlib • Seaborn • APIs & Web Scraping • IBM Cognos Analytics • JupyterLab